RUBY LASER

Ruby laser Introduction:-

- A ruby laser is a solid-state laser that uses the synthetic ruby crystal as its laser medium. Ruby laser is the first successful laser developed by Maiman in 1960.
- Ruby laser is one of the few solid-state lasers that produce visible light. It emits deep red light of wavelength 694.3 nm.

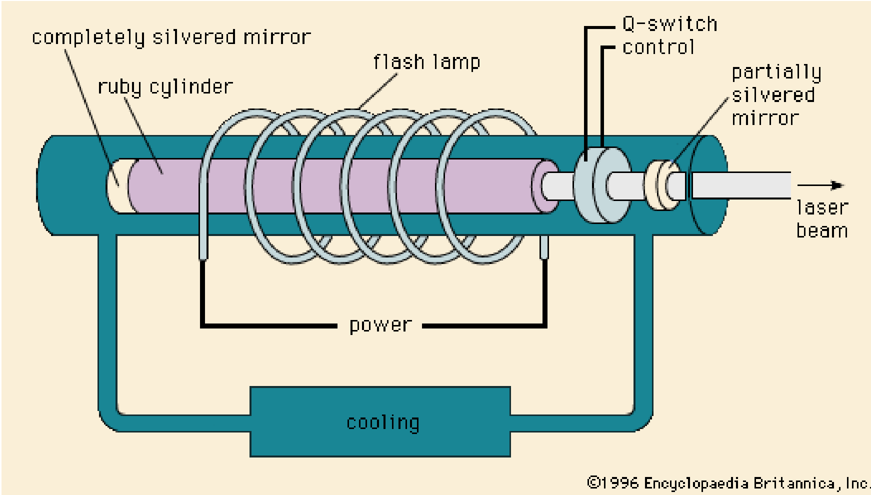


Figure:- Schematic diagram of Ruby Laser

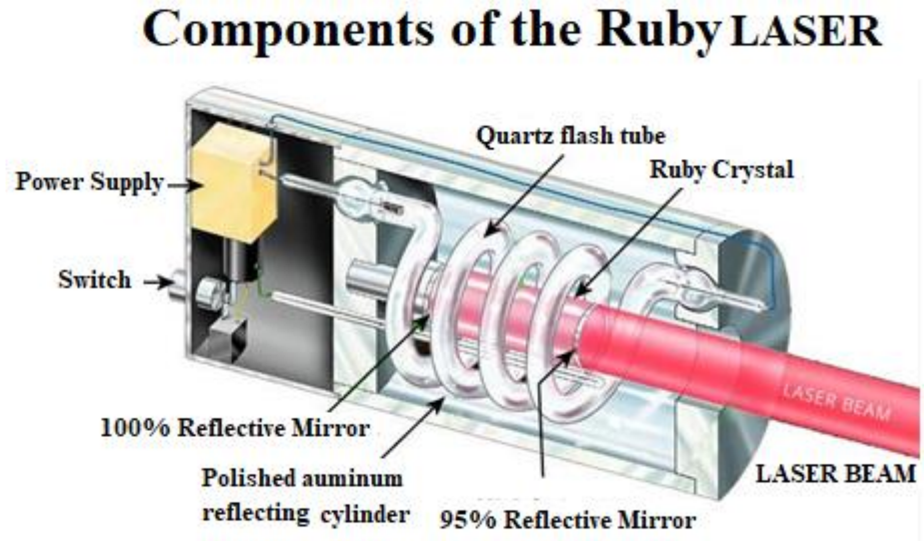


Figure:- Experimental setup of Ruby Laser

Construction of ruby laser:-

A ruby laser consists of three important elements: laser medium, the pump source, and the optical resonator.

(i) Active Medium:-

In a ruby laser, a single crystal of ruby [Ruby crystal is an Aluminium Oxide (Al_2O_3)] in the form of cylinder acts as a laser medium or active medium. It consists of a pink ruby cylindrical rod about 5 cm in length and 1 cm in diameter.

The laser medium (ruby) in the ruby laser is made of the host of Sapphire (Al_2O_3) which is doped with small amounts (0.05% by weight) of chromium ions (Cr^{3+}).

The ends of the rod are made optically plane and exactly parallel to each other.

The ends are silver polished so that one of them is fully silvered While the other one is partially reflecting for the emergence of laser light beam

(iii) Pump source or energy source in ruby laser:-

The pump source is the element of a ruby laser system that provides energy to the laser medium. In a ruby laser, population inversion is required to achieve laser emission. Population inversion is the process of achieving the greater population of atoms at higher energy state than the lower energy state. In order to achieve population inversion, we need to supply energy to the laser medium (ruby rod).

In a ruby laser, we use Xenon flashtube as the energy source or pumping source. The flashtube supplies energy to the laser medium (ruby). When atoms of lower energy state in the laser medium gain sufficient energy from the flashtube, they jump into the higher energy state or excited state.

(iv) Optical resonator:-

The ends of the cylindrical ruby rod are flat and parallel. The cylindrical ruby rod is placed between two mirrors. The optical coating is applied to both the mirrors. The process of depositing thin layers of metals on glass substrates to make mirror surfaces is called silvering. Each mirror is coated or silvered differently.

At one end of the rod, the mirror is fully silvered whereas, at another end, the mirror is partially silvered.

The fully silvered mirror will completely reflect the light whereas the partially silvered mirror will reflect most of the part of the light but allows a small portion of light through it to produce output laser light.

Working of ruby laser:-

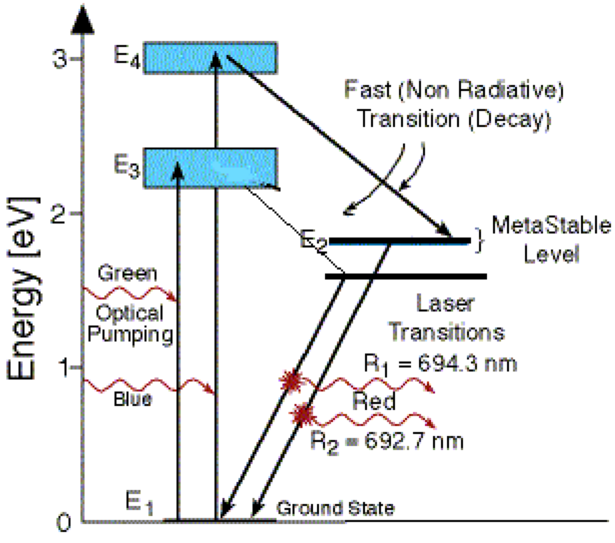
The ruby laser is a three level solid-state laser. In a ruby laser, **optical pumping technique** is used to supply energy to the laser medium. Optical pumping is a technique in which light is used as energy source to raise electrons/Atoms from lower energy level to the higher energy level.

Consider a ruby laser medium consisting of three energy levels E_1, E_2, E_3 with N number of atoms.

We assume that the energy levels will be $E_1 < E_2 < E_3$. The energy level E_1 is known as ground state or lower energy state, the energy level E_2 is known as metastable state, and the energy level E_3 is known as pump state.

Let us assume that initially most of the electrons are in the lower energy state (E_1) and only a tiny number of electrons are in the excited states (E_2 and E_3).

When light energy is supplied to the laser medium (ruby), the electrons in the lower energy state or ground state (E_1) gains enough energy and jumps into the pump state (E_3).



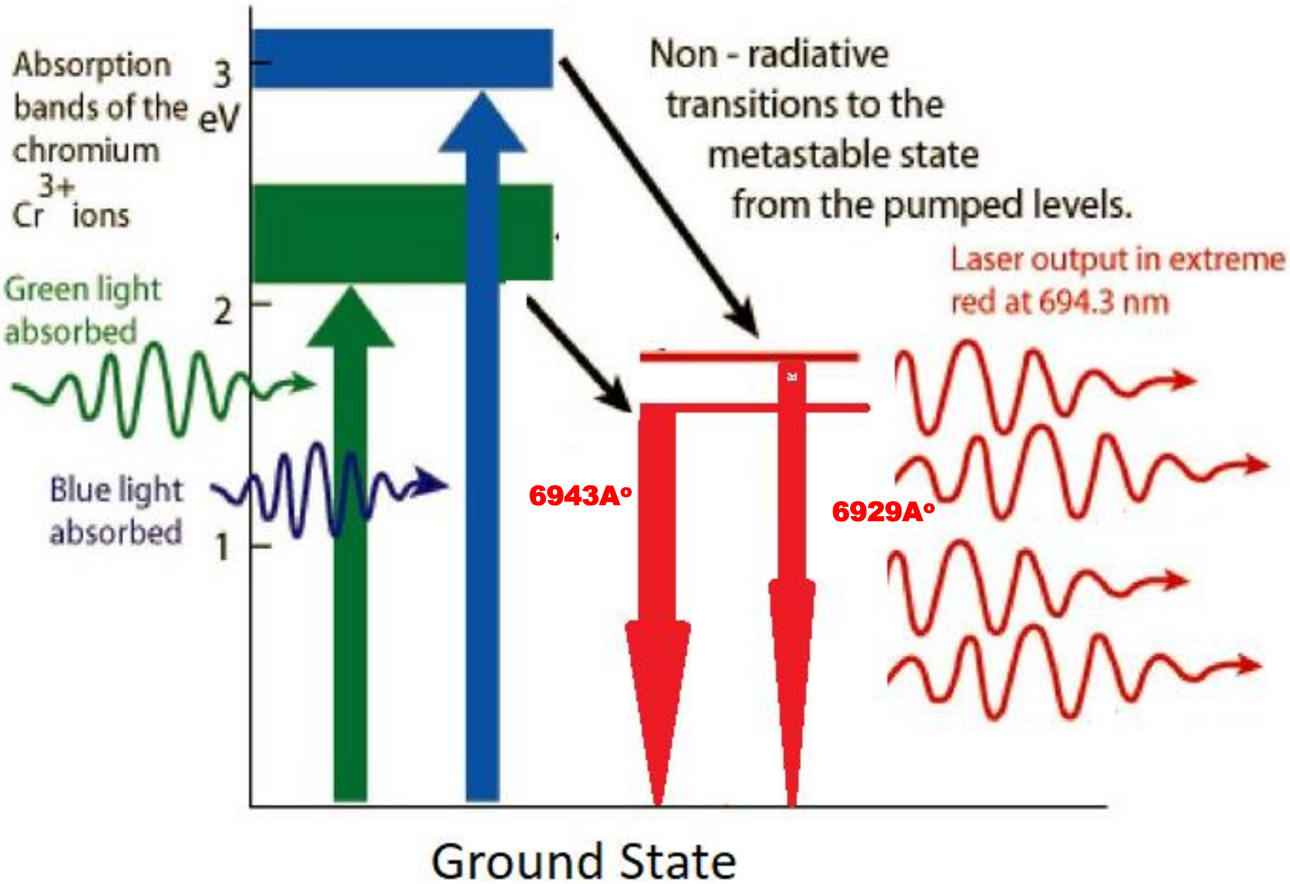
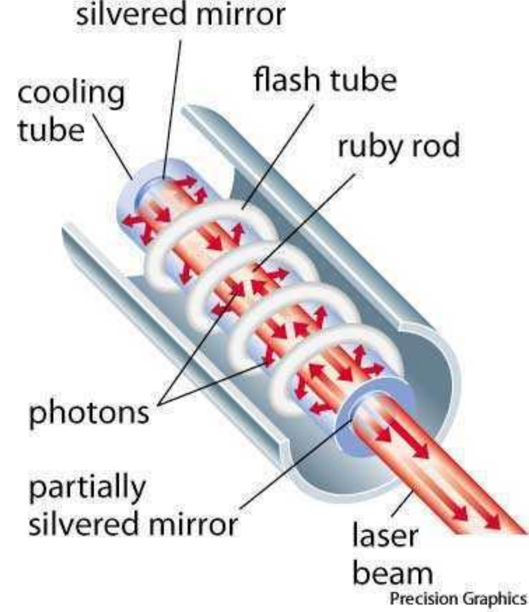
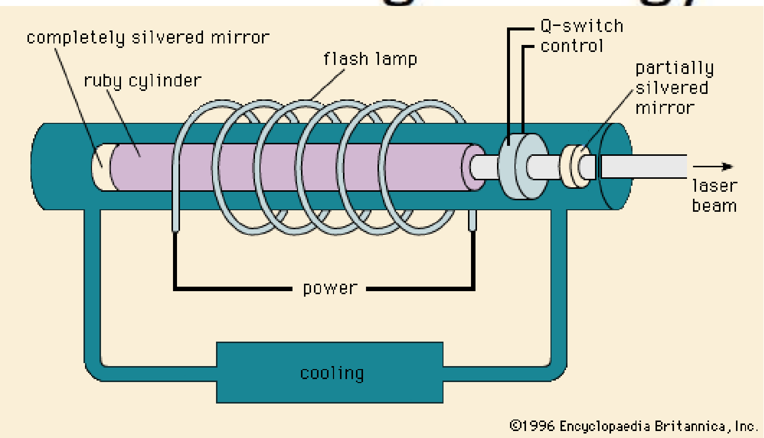
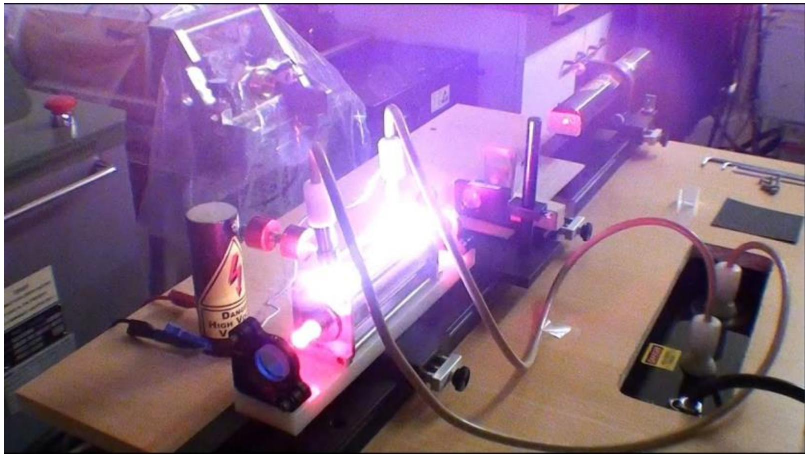


Fig.:- Energy levels of Cr^{3+}





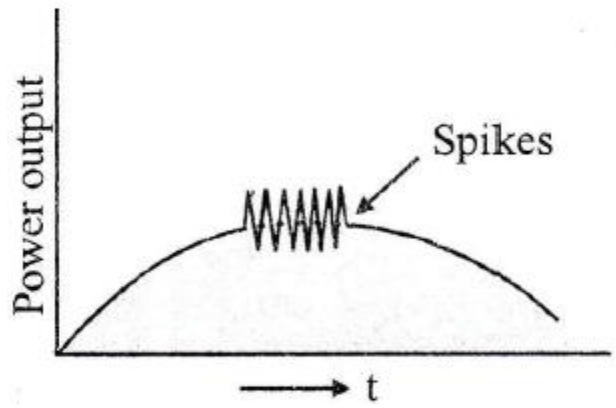


The lifetime of pump state E_3 is very small (10^{-8} sec) so the electrons in the pump state do not stay for long period. After a short period, they fall into the metastable state E_2 by releasing radiationless energy. The lifetime of metastable state E_2 is 10^{-3} sec which is much greater than the lifetime of pump state E_3. Therefore, the electrons reach E_2 much faster than they leave E_2. This results in an increase in the number of electrons in the metastable state E_2 and hence population inversion is achieved.
After some period, the electrons in the metastable state E_2 falls into the lower energy state E_1 by releasing energy in the form of photons. This is called spontaneous emission of radiation.
When the emitted photon interacts with the electron in the metastable state, it forcefully makes that electron fall into the ground state E_1. As a result, two photons are emitted. This is called stimulated emission of radiation.

When these emitted photons again interacted with the metastable state electrons, then 4 photons are produced. Because of this continuous interaction with the electrons, millions of photons are produced.
In an active medium (ruby), a process called spontaneous emission produces light. The light produced within the laser medium will bounce back and forth between the two mirrors. This stimulates other electrons to fall into the ground state by releasing light energy. This is called stimulated emission. Likewise, millions of electrons are stimulated to emit light. Thus, the light gain is achieved.
The amplified light escapes through the partially reflecting mirror to produce laser light.

SPIKING IN RUBY LASER

Ruby laser produces high energy pulses of coherent light of very short duration of the order of microsecond. These pulses are called spikes as shown in Fig.



When the pulsating flash lamp is suddenly switched on, the atoms in the lower state get energy and some of these atoms transit to the metastable state. As soon as the energy given by the flash lamp reaches a threshold value (a minimum level of energy needed to produce laser action), the population inversion builds up and the laser process sets in. This in turn depletes the population of the metastable state below the threshold value. Consequently the laser action ceases for a few micro seconds. As the flash lamp continues to excite the atoms to the metastable state again laser action sets in. This lasing action repeats itself till the flash lamp power falls below the threshold value and lasing action stops.

Advantages and Disadvantage

- **Benefits or advantages of Ruby Laser**

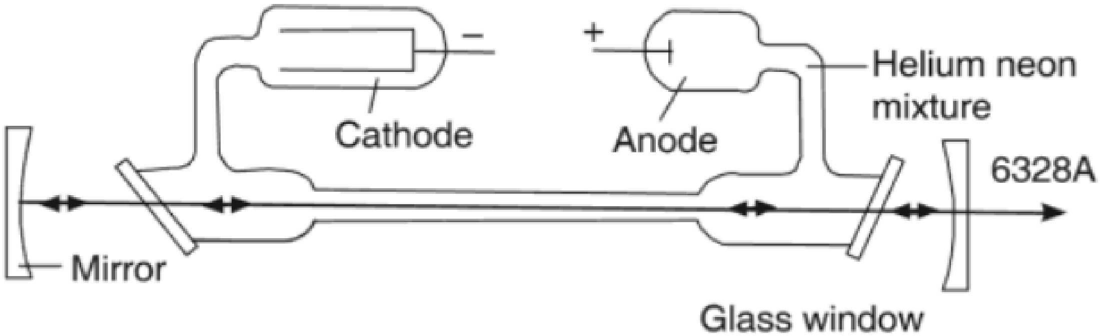
➡They are economical.
➡Beam diameter of ruby laser is comparatively less than CO_2 laser type.
➡Output power of ruby laser is not as less as He-Ne laser type.
➡Ruby is in solid form and hence there is no chance of wasting material of the active medium.
➡Due to their low output power, they are known as class-I lasers. Hence they are used as toys for children. They can also be used as decoration piece and artistic display.

- **Drawbacks or disadvantages of Ruby Laser**

➡No significant stimulated emission occurs in ruby laser until at least half of the ground state electrons have been excited to the meta stable state.
➡Efficiency of this laser type is comparatively lower.
➡Optical cavity of this laser is short as compare to other laser types.

He-Ne Laser

The first gas laser to be operated successfully was the He-Ne laser fabricated by Ali Javan and his co-workers of the Bell telephone Laboratories in USA. The active medium in this laser in a mixture of He and Ne gas at low pressure. Lasing action takes place between specific levels of Neon. Pumping is provided by means of an electric discharge through the gas mixture.



(a) Construction

➢It consists of a Gas Discharge Tube made of quartz or borosilicate glass about 40cm long and 2 mm in diameter with parallel mirrors at both ends. One of them is 100% reflecting and the other is partially transparent.

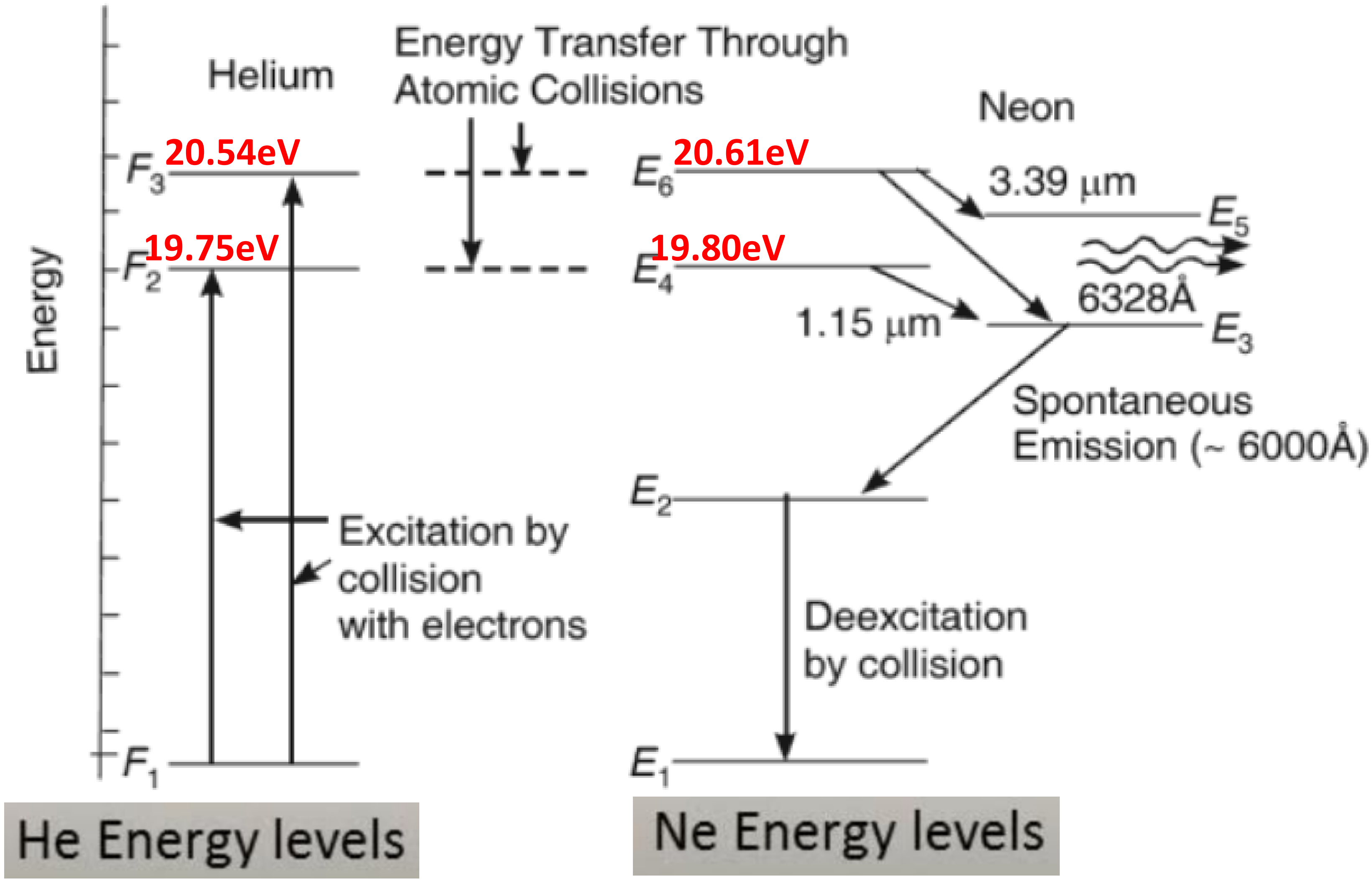
➢A mixture of about 10 parts of helium and 1 part of neon at low pressure [approximately - 1 Torr (1mm of mercury)] is placed in the discharge tube.

➢The energy or Pumping Source of the laser is provided by an electrical discharge of around 1000 volts through an anode and cathode at each end of the glass tube.

➢The Brewster's windows of the discharge tube ensure that the beam is linearly polarized

➢The Optical Resonant Cavity of the laser typically consists of a plane, high- reflecting mirror at one end of the laser tube, and a concave output coupler mirror of approximately 1% transmission at the other end.

(b) Energy Level Diagram of a He-Ne laser



He-Ne laser employs a four level pumping scheme. The energy level diagram is shown in figure.

(C) Working

He-Ne laser employs a four level pumping scheme. The energy level diagram is shown in figure.

(i) Electrical Discharge Pumping:- When the power is switched on, the electric field ionizes some of the atoms in the mixture of helium and neon gases. Due to the electric field the electrons and ions will be accelerated towards the anode and the cathode respectively. Since the electrons have a smaller mass they acquire a high velocity. Through the collisions of energetic electrons, the helium atoms get excited to the upper levels and F3.

Helium atoms are more readily excitable than neon atoms because they are lighter. These two states are metastable state.

Though a radiative transition is forbidden, the excited helium atoms can return to the ground state by transferring their excess energy to the neon atoms through collisions. Such an energy transfer can take place when the two colliding atoms have nearly identical energy states.

The E 6 and E4 levels of neon nearly coincide in energy with the F2 and F3 levels of helium. The helium atoms de-excite to the ground state by imparting energy to neon atoms. This is the main pumping mechanism in He-Ne system.

(ii) Population Inversion & Stimulated Emission :-

Neon atoms are the active centres and the role of helium is to excite the neon atoms and cause population inversion. **The probability of energy transfer from helium atoms to neon atoms is more, as there are 10 helium atoms to 1 neon atom in the mixture. For this reason, the probability of reverse transfer i.e., from neon to helium is small.**

At ordinary temperatures the E5 and E3 (lower lasing levels) levels of neon are sparsely populated and a state of **population inversion** is achieved between E6, E5 and E4, E3. three laser transitions can occur. They are E6 —> E5, E6 —> E3 and E4—> E3 transitions.

As the terminal levels of lasing transitions are sparsely populated, the fraction of neon atoms that must be excited to the upper levels can be much less. As such power required for pumping is low, random photons emitted spontaneously set on stimulated emissions and coherent radiation is produced.

E6 —> E3 transition: This transition generates a laser beam of wave length 6328Å (Red colour).
E4 —> E3 transition: This produces an infra- red laser beam of wavelength 11500Å.
E6 —>E5, transition: A laser beam of wave length 33900 Å in the far-infra red region arises due to this transition.

From the terminal levels E, and E3 of lasing transitions neon atoms can make downward transitions to E2 level. The E2 level is a metastable state. Therefore neon atoms tend to accumulate at this level. These atoms collide with the walls of the tube and impart their energy and return to the ground state. These atoms are to be brought to the ground state quickly; otherwise the number of atoms available at the ground state will go on diminishing and the laser output decreases.

To increase the probability of atomic collisions with the walls, the discharge tube is made very narrow.
This transition (E6 —> E3 transition) generates a laser beam of wave length 6328Å
It falls on **Brewster window** at Brewster’s angle, thereby producing a completely polarized light.
The high directivity of a laser beam is achieved by the parallel mirrors fixed at the end of the and the **resonator cavity** made by them.

The efficiency of He-Ne laser is less. However, the laser beam from the gas laser is more monochromatic and directional (beam spread is less) than solid state lasers.

Gas lasers are capable of operating continuously and do not need cooling. The power output is however much smaller, 1-50 mw with an input of 5 to 10 W.

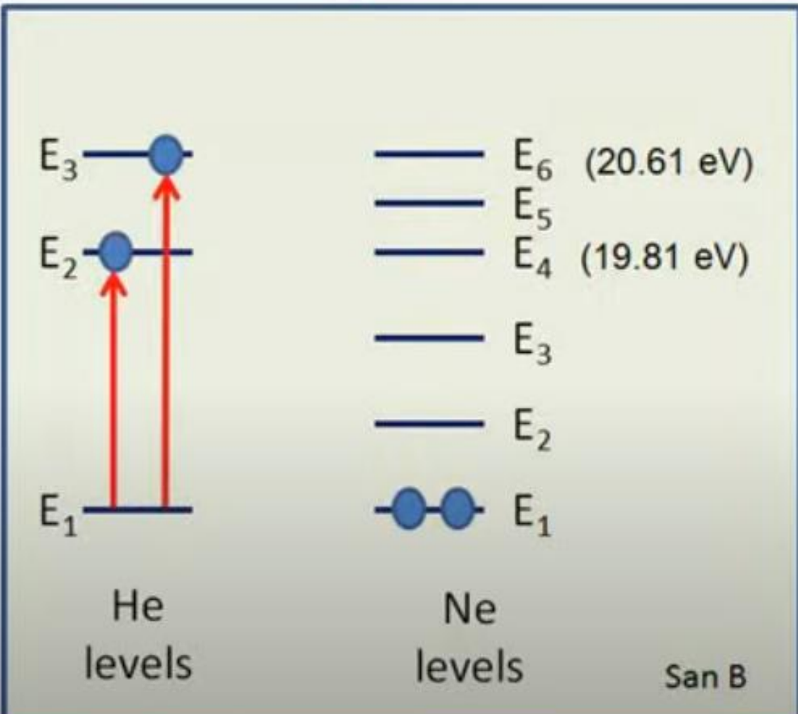
Advantages

- They are small and compact.
- They offer best inherent beam quality of any laser which virtually pure single transverse mode beam (M2 < 1.05).
- They have longer life, usually 50,000 hours or more.
- They generate relatively little heat. Moreover they are convection cooled easily in OEM packages.
- They have relatively low acquisition and operating cost.
- Construction of He-Ne laser is very simple.
- It has very good coherence property.
- It provides inherent safety.
- He Ne Laser is produced Plane polarized Laser Light

Disadvantages

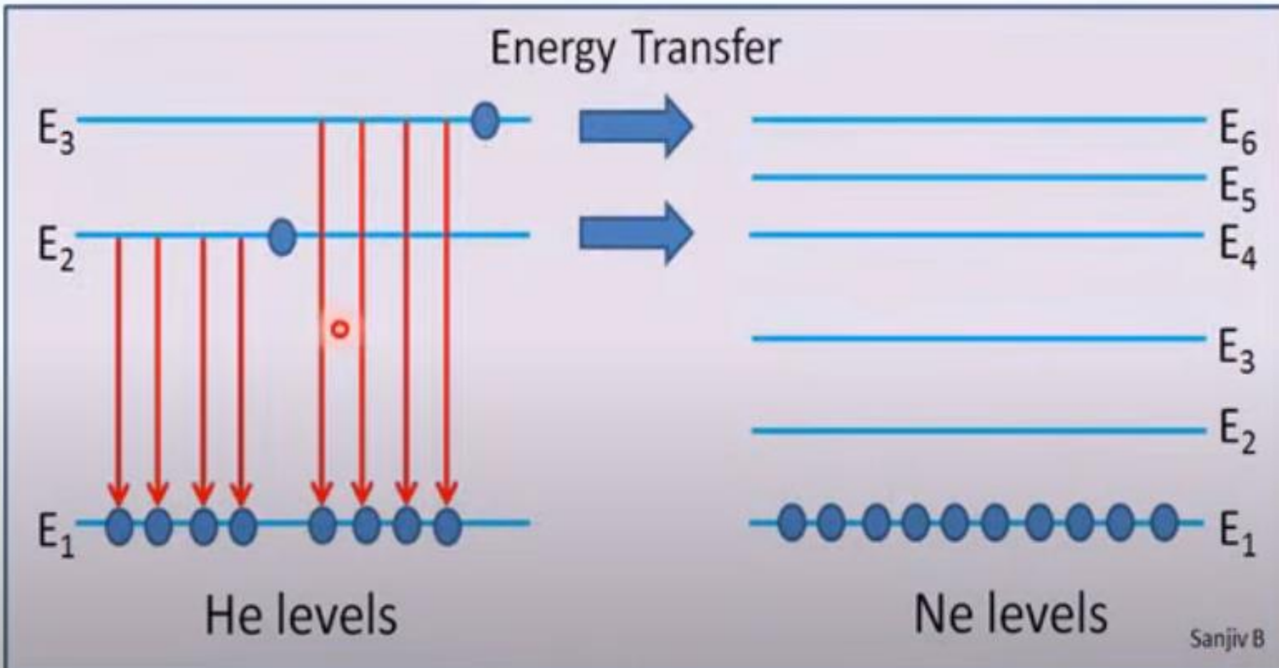
- Its output power is lower.
- It is a low gain device.
- In order to have operation at single wavelength, the other two wavelengths are required to be suppressed. This requires use of special techniques and extraordinary skills. This increases cost of the device.
- It requires high voltage for its operation.
- Escaping of gas from laser plasma is considered to its drawback.

Role of Helium in He Ne LASER



- The helium atoms are lighter. So more readily excitable than neon atoms.
- Helium atoms in excited energy levels E_2 and E_3 collide with the Neon atoms in the ground level. Neon atoms are excited to energy levels E_4 and E_6 and Helium atoms come back to the ground state.
- **The neon atoms are much heavier and could not be pumped efficiently without Helium atoms.**
- **The role of Helium atoms is to excite Neon atoms and cause population inversion.**

Why the proportion of He : Ne is 10:1 ?



- Many He atoms excited to higher energy levels E2 and E3 will fall down due to spontaneous emission and wont be able to transfer their energy to Ne atoms.
- The probability of energy transfer from Helium atoms to Neon atoms is more as there are 10 Helium atoms per 1 Neon atom in a gas mixture.

The probability of energy transfer from helium atoms to neon atoms is more, as there are 10 helium atoms to 1 neon atom in the mixture. For this reason, the probability of reverse transfer i.e., from neon to helium is small.

Why is helium gas used in the helium neon laser system?

The role of the Helium gas in He-Ne laser is to increase the efficiency of the lasing process. Two effects make Helium particularly valuable:

1.The direct excitation of Neon gas is inefficient, but the direct excitation of He gas atoms is very efficient.

2. The Helium atoms have a lower energy threshold for excitation than Ne atoms

3.An excited state of the He atom (labeled F2 and F3) has very similar to the energy of an excited state of the Neon atom (also labeled E4 and E6).

The excitation process of the Neon atoms is a **two stages process:**

•The high voltage causes electrons to accelerate from the cathode toward the anode. These electrons collide with the He atoms and transfer kinetic energy to them.

•The excited Helium atoms collide with the Neon atoms, and transfer to them the energy for excitation.

What is the function of the He atoms in the He-Ne laser?

In the He-Ne laser, the helium atoms are used to pump the neon atoms to a higher energy level, and then the neon atoms release the energy in the form of coherent light. The helium atoms do not directly emit laser light in the He-Ne laser, but they play an important role in the lasing process.

Why helium atoms have a lower energy threshold for excitation than Ne atoms?

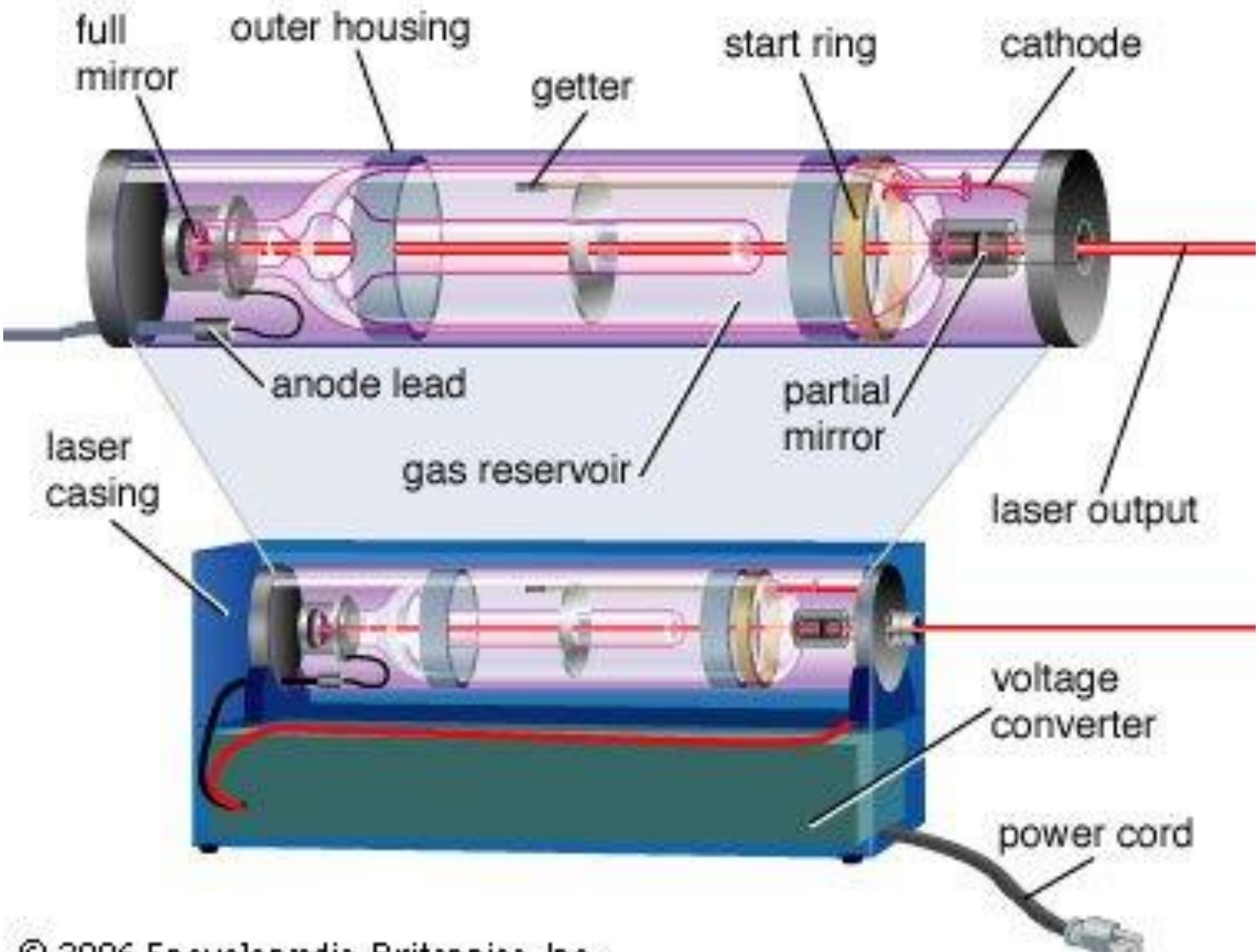
The energy thresholds for the excitation of atoms, such as helium (He) and neon (Ne), are influenced by their electronic structure. The reason helium atoms have lower energy thresholds for excitation than neon atoms can be understood through a few key points related to atomic structure and quantum mechanics:

Atomic Structure: Helium has a simpler atomic structure with only two electrons in the 1s orbital, while neon has a more complex structure, with ten electrons filling the 1s, 2s, and 2p orbitals.

Quantum Mechanical Principles: The quantum mechanical model of the atom explains that the energy levels are not only quantized but also depend on the quantum numbers that describe the electrons' states. The larger the gap between the energy levels, the more energy is required to excite an electron from one level to another.
In helium, the first excitation energy is relatively low because moving an electron to the next available energy level (for example, from the 1s to the 2p level, since helium does not naturally have electrons in the 2s level due to its electron configuration) requires less energy compared to neon, where electrons are already in higher energy states (2s and 2p) and moving them to even higher levels (e.g., 3s or 3p) requires more energy.

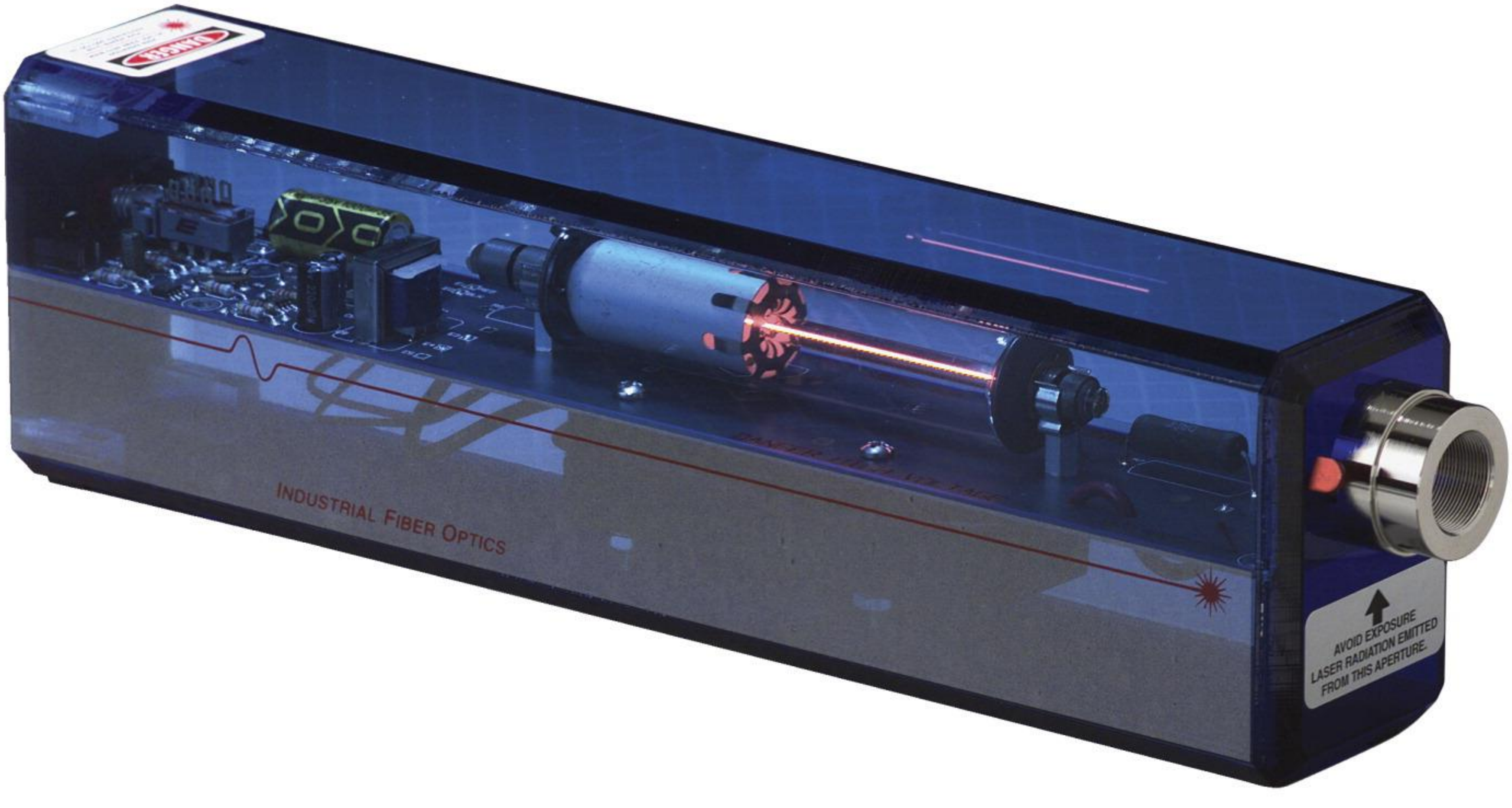
Difference Between Ruby Laser & He-Ne-Laser

	Ruby Laser	He-Ne-Laser
Property	It produces pulsating laser	It produces Continuous laser
Energy Level	It is a three energy level system	It is a four energy level system
Pumping	Optical pumping produces population inversion	Electric Discharge produces population inversion
Cooling	Cooling arrangement is required	Cooling arrangement is not required
Collimated	Laser light from this laser is less collimated and coherent.	Laser light from this laser is highly collimated and coherent.
Wavelength	It emits light of wavelength 6943 Å	It emits light of wavelength 6328 Å
Power	Power output 10 to 20KW	Power output 1 to 50W
Polarization	This is Un-polarized laser	This is linearly polarized laser


full mirror
outer housing
getter
start ring
cathode
anode lead
partial mirror
laser casing
gas reservoir
laser output
voltage converter
power cord

Gas Discharge Tube

Complete Unit of He-Ne Laser



LASER APPLICATIONS

- **In Surgery:** Lasers are used to perform bloodless surgery of tumours, retina, etc.
- **In Engineering:** Strong laser beams are used to make accurate welds.
- **In Communication:** Lasers are used to send signals over large distances through optical fibres.
- **In industry:** Lasers are used for drilling holes in diamonds and hard steel. When focussed through a lens, they can melt and vaporise metals.
- **In Weather forecasting:** A laser-based radar called Lidar is used for weather forecasting.
- **For Defence purposes:** Laser beams are used for controlling and guiding rockets, missiles, satellites and location of aircrafts. Intense laser beams can even destroy tanks and planes.
- **In Holography:** Laser beams are used for a three-dimensional photography without lenses, called holography.
- **In computers:** Laser beams are being used to transmit an entire memory bank from one computer to another.

Details of Applications of Lasers

- ✓ Laser is an optical device that generates intense beam of coherent monochromatic light by stimulated emission of radiation.
- ✓ Laser light is different from an ordinary light. It has various unique properties such as coherence, monochromacity, directionality, and high intensity. Because of these unique properties, lasers are used in various applications.
- ✓ **The most significant applications of lasers include:-**
  - ➢ Lasers in medicine
  - ➢ Lasers in communications
  - ➢ Lasers in industries
  - ➢ Lasers in science and technology
  - ➢ Lasers in military

Continue………

LASERS IN MEDICINE

- Lasers are used for bloodless surgery.
- Lasers are used to destroy kidney stones.
- Lasers are used in cancer diagnosis and therapy.
- Lasers are used for eye lens curvature corrections.
- Lasers are used in fiber-optic endoscope to detect ulcers in the intestines.
- The liver and lung diseases could be treated by using lasers.
- Lasers are used to study the internal structure of microorganisms and cells.
- Lasers are used to produce chemical reactions.
- Lasers are used to create plasma.
- Lasers are used to remove tumors successfully.
- Lasers are used to remove the caries or decayed portion of the teeth.
- Lasers are used in cosmetic treatments such as acne treatment, cellulite and hair removal.

LASERS IN COMMUNICATIONS

- Laser light is used in optical fiber communications to send information over large distances with low loss.
- Laser light is used in underwater communication networks.
- Lasers are used in space communication, radars and satellites.

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LASERS IN INDUSTRIES

- Lasers are used to cut glass and quartz.
- Lasers are used in electronic industries for trimming the components of Integrated Circuits (ICs).
- Lasers are used for heat treatment in the automotive industry.
- Laser light is used to collect the information about the prefixed prices of various products in shops and business establishments from the bar code printed on the product.
- Ultraviolet lasers are used in the semiconductor industries for photolithography. Photolithography is the method used for manufacturing printed circuit board (PCB) and microprocessor by using ultraviolet light.
- Lasers are used to drill aerosol nozzles and control orifices within the required precision.

Continue………

Lasers in Science and Technology

- A laser helps in studying the Brownian motion of particles.
- With the help of a helium-neon laser, it was proved that the velocity of light is same in all directions.
- With the help of a laser, it is possible to count the number of atoms in a substance.
- Lasers are used in computers to retrieve stored information from a Compact Disc (CD).
- Lasers are used to store large amount of information or data in CD-ROM.
- Lasers are used to measure the pollutant gases and other contaminants of the atmosphere.
- Lasers helps in determining the rate of rotation of the earth accurately.
- Lasers are used in computer printers.
- Lasers are used for producing three-dimensional pictures in space without the use of lens.
- Lasers are used for detecting earthquakes and underwater nuclear blasts.
- A gallium arsenide diode laser can be used to setup an invisible fence to protect an area.

Continue………

Lasers in Military

- Laser range finders are used to determine the distance to an object.
- The ring laser gyroscope is used for sensing and measuring very small angle of rotation of the moving objects.
- Lasers can be used as a secretive illuminators for reconnaissance during night with high precision.
- Lasers are used to dispose the energy of a warhead by damaging the missile.
- Laser light is used in LIDAR's to accurately measure the distance to an object.

FREQUENTLY ASKED QUESTIONS (FAQS)

Q. In laser why the atom is not directly excited to a metastable state instead of an excited state?

Ans:- Based on quantum mechanics principles it is a lot easier to excite the atom to a higher level, relatively more stable, excited state than a metastable state.

Metastable states has a long lifetime (hence the name) since the transitions to the ground state are "not allowed" when considering the different transition rules, The probability for a transition (Einstein A and B) are much smaller which mean that if we try and populate the metastable states directly with lasers the prbability is quite low, and a population using a excited state will be more efficient in almost all possible cases.

Note that this does not apply for non-radiative excitation as in the case of electric discharge or energy transfer as is the case in a He-Ne where Ne is exited via the excited state in He which is turn is excited by the electric discharge.

Q. What is optical pumping?

Ans:- In optical pumping a light source generates photons with enough energy that they are able to get absorbed by the atoms in the lasing medium and cause them to go into an excited state.

Q. What is the lifetime of that state?

Ans:- The time that it spends, on average, in that excited level is called the lifetime of that state. Lifetimes can vary in duration depending on the atom and on the energy level. Excited state lifetimes are typically a few nanoseconds (10^{-9} s, or a billionth of a second), but they could be as short as a picosecond (10^{-12} s, or a thousand times shorter still) or as long as a few milliseconds (10^{-3} s). The ground state, of course, has an infinitely long lifetime since an atom in its ground state can no longer decrease its energy. So, the most stable of states is the ground state. Long-lived states are referred to as meta-stable states.